

The Skin and Systemic Disease: A Comprehensive Study Guide

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■ Learning Objectives

- Recognize **cutaneous clues** that point toward underlying systemic disease
- Identify **characteristic rashes** that indicate specific systemic conditions
- Understand **skin reactions associated with infections** (toxic erythema, EM, EN, EAC, ECM)
- Correlate **skin changes with hormonal, GI, liver, and malignant diseases**
- Differentiate **hypopigmentation and hyperpigmentation disorders**
- Recognize **dermatological manifestations of pregnancy**

■ Key Takeaways

- > 1. **The skin is a window to systemic disease** — rashes with joint pain, fever, weight loss, SOB, or altered bowel function should always raise suspicion of an underlying systemic condition.
- > 2. **Non-blanching palpable purpuric lesions = vasculitis** until proven otherwise; think Hep B/C, SLE, lymphoma, leukemia.
- > 3. **Specific rash–disease associations** are high-yield: Erythema multiforme → HSV; Pyoderma gangrenosum → IBD/RA/malignancy; Erythema nodosum → Strep/TB/sarcoidosis/IBD/Behçet's.
- > 4. **Skin changes in hormonal disease** follow patterns: Addison's → generalized hyperpigmentation; Acanthosis nigricans → insulin resistance/gastric cancer; Necrobiosis lipoidica → DM.
- > 5. **Pregnancy dermatoses** range from benign (Prurigo, PEP) to serious (Pemphigoid gestationis — autoimmune, affects baby, recurs).

■ Main Content

■ Clues That a Rash Indicates Systemic Disease

When evaluating a skin rash, the following features should raise a **red flag** for underlying systemic disease:

- ■ Rash **associated with** joint pain, fever, weight loss, weakness, shortness of breath, or altered bowel function
- ■ Rash **not responding to topical treatment**
- ■ **Erythema** (inflammation around blood vessels)
- ■ **Non-blanching, palpable, purplish, fixed, painful, blistering lesion** → Think **VASCULITIS**
- ■ **Unusual changes in pigmentation or texture**
- ■ **Palpable dermal lesion** → secondary to granuloma, metastasis, lymphoma, etc.

> ■ **Key Insight:** A rash that is widespread, symmetric, non-blanching, or associated with systemic symptoms is rarely just a dermatological problem — always investigate for systemic causes.

■ Characteristic Rashes Indicating Systemic Disease

| Rash | Associated Systemic Diseases |

|---|---|

| **Erythema multiforme** | HSV (most common), Hep B/C, Mycoplasma pneumonia |

| **Pyoderma gangrenosum** | IBD (UC, Crohn's), RA, hematological malignancy |

| **Erythema nodosum** | IBD, Streptococcal infection, TB, Sarcoidosis, Behçet's |

| **Vasculitis** | Hep B/C, SLE, Lymphoma, Leukemia |

> ■ **Key Insight:** These four rashes are the "classic four" that every medical student must memorize. They are the skin's way of screaming that something systemic is wrong.

■ Skin Reactions Associated with Infections

1. ■ Toxic Erythema

- **Presentation:** Widespread, symmetrical **maculopapular erythema** (morbilliform — "measles-like")
 - **Distribution:** Starts on the **trunk**, then spreads to the **limbs**
 - **Symptoms:** Usually **asymptomatic** (may be itchy)
 - **Blanching:** Yes (blanching)
 - **Triggers:** Viruses, bacteria, **drugs** (most common cause in clinical practice)
 - **Classic presentation:** Drug-induced morbilliform eruption (e.g., ampicillin in EBV infection)
 - **Treatment:** Usually **no treatment needed**
 - Emollients and topical steroids if symptomatic
 - **Treat the underlying cause** (infection or discontinue offending drug)
- > ■ **Quick Fact:** The most common cause of toxic erythema in clinical practice is **drug reactions**, especially antibiotics (ampicillin, sulfonamides).

2. ■ Erythema Multiforme (EM)

Epidemiology:

- Most common in **young adults**
- **Males > Females**
- Most common infectious trigger: **HSV** (Herpes Simplex Virus)

Presentation:

- **Target lesions** (also called "iris lesions") — the hallmark
- **3 zones/colors:** Dusky red center (necrosis) → pale edematous ring → outer erythematous ring
- Resembles a "bull's eye" or **"doll's eye"**
- **Acral distribution** (hands, feet, extremities)
- Systemic symptoms of the underlying infection (e.g., HSV) usually **precede the rash by 2–14 days**

Pathophysiology:

- **Type IV Hypersensitivity Reaction** (cell-mediated, delayed-type)

Classification:

Feature	EM Minor	EM Major
Skin involvement	Target lesions, acral	Target lesions + mucosal involvement
Mucosal involvement	Minimal or absent	Significant (oral, genital, conjunctival)

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| Skin involvement | Target lesions, acral | Target lesions + mucosal involvement |

| Mucosal involvement | Minimal or absent | Significant (oral, genital, conjunctival) |

| Systemic symptoms | Mild | Severe (fever, malaise) |

| Example cause | HSV | Mycoplasma, drugs |

****Diagnosis:** Clinical**** — based on characteristic target lesions and history

****Treatment:****

- EM Minor: Supportive, treat underlying cause (e.g., acyclovir for HSV)
- EM Major: May require systemic steroids (controversial), supportive care

****Recurrence:****

- ****Yes**** — especially if HSV-triggered (recurrent HSV → recurrent EM)
- Prophylactic acyclovir may be used for recurrent cases

****Course:**** Self-limiting, resolves in ****1–2 weeks****

****Rare associations:**** Hepatitis B/C, Mycoplasma pneumoniae, drugs (sulfonamides, phenytoin)

> ■ ****Key Insight:**** EM is essentially the skin's reaction to an internal trigger. ****Always look for the cause**** — HSV is #1, but drugs, mycoplasma, and hepatitis are also important.

> ■ ****Mnemonic for EM target lesion:** "Doll's Eye"** — Dusky center, Outer pink ring, Lighter middle zone = 3 zones

3. ■ Erythema Nodosum (EN)

****Definition:**** The ****most common form of panniculitis**** (inflammation of subcutaneous fat tissue)

****Epidemiology:****

- ****Females >> Males**** (especially young women, 20–30 years)
- Most common panniculitis overall

****Presentation:****

- ****Acute, painful, tender nodules**** — especially on the ****shins**** (bilateral)
- Nodules are ****3–20 cm****, ****not sharply marginated****
- Very tender on palpation
- Overlying skin is erythematous and warm

****Etiology:****

| Category | Causes |

|---|---|

| ****Idiopathic**** | 50% of cases |

| ****Infections**** | Streptococcal (most common secondary cause), TB, Yersinia, Schistosomiasis |

| ****Inflammatory**** | IBD, Sarcoidosis, Behçet's disease |

| **Drugs** | OCPs, Sulfonamides |

| **Malignancy** | Lymphoma (rare) |

> ■ **Mnemonic for EN causes — "STEPS TO BIS" (modified):**

> **S**treptococcal, **T**B, **E**nteric (Yersinia), **P**regnancy/OCPs, **S**arcoidosis, **T**B, **O**ther (Behçet's), **B**owel (IBD), **I**diopathic, **S**chistosomiasis

Diagnosis & Management:

- Bacterial culture (throat swab for Strep)
- Imaging (CXR for sarcoidosis, TB)
- Dermatopathology: **Septal panniculitis without vasculitis**

Course: **Spontaneous resolution in ~6 weeks**

Treatment:

- Treat the underlying cause
- **Rest + NSAIDs** for pain
- Steroids: controversial, used in severe cases
- Potassium iodide (alternative)

> ■ **Key Insight:** EN is a **reactive process** — it's never the primary disease. Always investigate for secondary causes, especially **Strep, TB, sarcoidosis, and IBD**.

4. ■ Erythema Annulare Centrifugum (EAC)

- **Presentation:** Annular (ring-shaped) erythematous lesions that **expand centrifugally** (outward)
- **Key feature:** "Trailing scale" — a thin line of scale on the inner edge of the advancing border
- **Differential diagnosis:** Must distinguish from **Tinea corporis** (ringworm)
- KOH preparation to rule out fungal infection
- **Associations:** Can be idiopathic or associated with underlying malignancy (especially in older adults), infections, or drugs

> ■ **Quick Fact:** EAC is sometimes called a "tumor marker rash" — in elderly patients with new-onset EAC, always consider underlying malignancy.

5. ■ Erythema Chronicum Migrans (ECM)

- **Cause:** **Borrelia burgdorferi** — the spirochete that causes **Lyme disease**
- **Transmission:** Tick bite (Ixodes tick)
- **Presentation:**

- **Migrating erythema** — a slowly expanding erythematous patch with central clearing ("bull's eye" or "target" appearance)
 - Develops at the site of the tick bite, typically **3–30 days** after the bite
 - May be accompanied by **flu-like symptoms** (fever, fatigue, headache, myalgias, arthralgias)
 - **Pathophysiology:** Cutaneous inflammatory response to the spirochete; hematogenous spread can lead to **CNS involvement** (meningitis, cranial nerve palsies — especially CN VII), cardiac involvement (AV block), and arthritis
 - **Treatment:** Doxycycline (first-line), amoxicillin, ceftriaxone
- > ■ **Key Insight:** ECM is the **earliest clinical sign of Lyme disease**. Recognizing it early and treating appropriately prevents serious cardiac and neurological complications.
- > ■ **Mnemonic for Lyme disease:** **"BORRELIA"** — **B**ull's eye rash, **O**rtho-arthritis, **R**emember tick bite, **R**ecurrent episodes, **E**rythema migrans, **L**ate arthritis, **I**mmune complex disease, **A**V block (cardiac)

■ Sarcoidosis

Etiology: Unknown — possibly triggered by **atypical mycobacterium** or other environmental antigens in genetically susceptible individuals

Key Features:

- **Multisystemic** granulomatous disease
- **Non-caseating granulomas** on histology (distinguishes from TB, which has caseating granulomas)
- **Bilateral hilar lymphadenopathy (BHL)** on CXR — classic finding

Skin Lesions — Specific vs. Non-specific:

Type	Lesion	Significance
Specific	(granulomatous)	Skin-colored papules (face) Earliest skin lesion; heal without scarring
	Plaques (annular, extremities, buttocks)	Chronic disease; SCARRING
	Lupus pernio	Violaceous plaques on nose and cheeks; bad prognostic sign
Non-specific	Erythema nodosum	Most common non-specific skin lesion; indicates early/benign sarcoidosis
Lupus Pernio:		
	Violaceous (dusky purple) plaques	on the nose, cheeks, and ears

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Non-specific	Erythema nodosum	Most common non-specific skin lesion; indicates early/benign sarcoidosis
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Lupus Pernio:

- **Violaceous (dusky purple) plaques** on the nose, cheeks, and ears

- **Bad prognostic sign** — associated with chronic, progressive disease
- Associated with **upper respiratory involvement**, **bone cysts** (osteolytic lesions), and **acral involvement**
- **DDx:** Rosacea — but lupus pernio has a **deeper violaceous hue** and more indurated texture
- Scarring Sarcoidosis:**
- **DDx:** Hypertrophic scar, pre-existing scar sarcoid (sarcoid can develop within old scars)
- Löfgren Syndrome (Acute Sarcoidosis):**
- > ■ **Mnemonic — "FLARE":**
- > **F**ever, **L**ymphadenopathy (bilateral hilar), **A**rthritis, **R**ash (Erythema nodosum), **E**rythema nodosum
- **Acute, multisystemic syndrome**
- **Benign** — resolves spontaneously
- Classic triad: **EN + Fever + Bilateral hilar lymphadenopathy** (+/- arthritis)
- Diagnosis:**
- **Imaging:** CXR (BHL), CT chest
- **Labs:** ↑ ACE levels, ↑ Calcium (hypercalcemia), ↑ 24-hour urinary calcium
- **Biopsy:** Non-caseating granulomas on lesional biopsy
- Treatment:**
- **Systemic involvement:** Systemic steroids (first-line)
- **Limited cutaneous disease:** Topical steroids
- **Extensive cutaneous disease:** Systemic steroids, **methotrexate**, **hydroxychloroquine**
- > ■ **Key Insight:** Sarcoidosis is a great mimicker — it can look like almost anything. The combination of **BHL + EN + hypercalcemia** is virtually diagnostic.

■ Skin Changes Associated with Hormonal Imbalance

Hyperpigmentation

- **Mechanism:** Increase in circulating hormones with **melanocyte-stimulating activity**
- **Causes:**
- **Addison's disease** (primary adrenal insufficiency) → ↑ ACTH → ↑ MSH → **generalized hyperpigmentation** (especially in creases, buccal mucosa, scars, sun-exposed areas)
- **Hyperthyroidism**, **Acromegaly**

- **Pregnancy, OCPs** → **Melasma/Chloasma** (localized hyperpigmentation on forehead and cheeks)

- **Treatment of melasma:** Sun protection (daily sunscreen) — may fade slowly if UV exposure is avoided

> ■ **Mnemonic for Addison's pigmentation:** **"ADDISON"** — **A**CTH **D**rives **D**arkening **I**n **S**kin **O**f **N**eed (adrenal insufficiency)

Hypopigmentation

- **Mechanism:** Widespread partial loss of melanocyte function

- **Cause:** **Hypopituitarism** — absence of MSH (melanocyte-stimulating hormone) → decreased pigmentation

Acanthosis Nigricans

- **Presentation:** Dark, velvety, thickened skin in **flexural areas** — axillae, groin, neck

- **Mechanism:** Increased thickness of epidermis due to **insulin resistance**

- **Associations:**

- **Obesity** (most common — benign type)

- **Insulin resistance / Type 2 DM**

- **Extensive involvement in older patients** → Think **GASTRIC ADENOCARCINOMA** (malignant acanthosis nigricans)

> ■ **Key Insight:** New-onset acanthosis nigricans in a **non-obese, older adult** should prompt a **malignancy workup**, especially for gastric adenocarcinoma.

Necrobiosis Lipoidica (NLD)

- **Association:** **Diabetes Mellitus** (but NOT related to severity of DM)

- **Presentation:**

- Well-demarcated, **yellowish-brown plaques** with **atrophic, translucent** center

- **Telangiectasia** visible through the thin skin

- Typically on the **pretibial area** (shins)

- **Course:** NOT affected by controlling blood sugar

- **Diagnosis:** Check **HbA1c** (65% of patients have or will develop DM)

- **Treatment:** Intralesional steroids, surgery (risk of ulceration)

> ■ **Quick Fact:** NLD can **precede** the diagnosis of diabetes by years — always check glucose metabolism in patients with this lesion.

Thyroid Disease

| Condition | Skin Manifestation |

|---|---|

| **Graves' disease** (hyperthyroidism) | **Pretibial myxedema** (dermopathy) — non-pitting, indurated plaques on shins; also ophthalmopathy, diffuse goiter |

| **Hypothyroidism** | **Generalized myxedema** — non-pitting edema, dry skin, coarse hair |

- **Pretibial myxedema treatment:** Topical/systemic steroids

> ■ **Key Insight:** Pretibial myxedema is specific to **Graves' disease** and is caused by glycosaminoglycan deposition in the dermis. It's a clue that the patient has autoimmune thyroid disease.

■ Skin Changes with GI and Liver Diseases

1. Zinc Deficiency

Acrodermatitis Enteropathica:

- **Genetic form:** Autosomal recessive disorder of zinc absorption (seen in **neonates**, especially after weaning from breast milk — breast milk contains zinc-binding ligands)

- **Acquired Zinc Deficiency (AZD):** Occurs in older individuals due to:

- Dietary deficiency

- Malabsorption (celiac disease, IBD, short bowel syndrome)

- Alcoholism

- Prolonged parenteral nutrition

Skin Changes:

- Appear **within weeks of birth** (genetic) or weeks after the precipitating cause (acquired)

- **Erythematous, inflamed, scaly skin** around **orifices** — mouth, anus, eyes (acral and periorificial distribution)

- If untreated → crusted, eroded, secondarily infected (Candida, S. aureus)

Other Findings:

- Diarrhea, alopecia, irritability, failure to thrive

Treatment: Zinc supplementation (1 mg/kg/day) — continued until levels normalize (lifelong in inherited forms)

> ■ **Key Insight:** The triad of **dermatitis + diarrhea + alopecia** in a neonate = think **acrodermatitis enteropathica** until proven otherwise.

2. Vitamin C Deficiency (Scurvy)

- **At-risk groups:** Malabsorption, poor diet, elderly, alcoholics

- **Mechanism:** Impaired collagen synthesis due to vitamin C deficiency → **vascular fragility**

****Skin Lesions:****

- ****Petechiae****, ****follicular hyperkeratosis**** with ****perifollicular hemorrhage**** (especially lower legs)
- ****Corkscrew hairs****, easy bleeding after teeth brushing, abnormal hair

****Other Findings:****

- Normocytic normochromic anemia
- Serum ascorbic acid level = zero
- X-ray findings are diagnostic (scorbutic rosary, Trümmerfeld zone)

****Treatment:**** Ascorbic acid (vitamin C) supplementation

> ■ ****Mnemonic for scurvy:**** **"**SCURVY**"** — ****S**welling gums**, ****C**orkscrew hairs**, ****U**nderperiosteal hemorrhage**, ****R**ough skin**, ****V**itamin C deficiency**, ****Y**ellowing (anemia)**

3. Pyoderma Gangrenosum (PG)

- ****Presentation:****
- ****Rapidly progressive, painful ulcer**** with ****necrotic base****
- ****Hypertrophic, undermined, purplish (violaceous) margins****
- Often triggered by minor trauma (****pathergy****) — e.g., tibial trauma → papule → rapidly enlarging ulcer
- ****Associations:****
- ****IBD**** (UC, Crohn's)
- ****Rheumatoid arthritis****
- ****Hematological malignancy**** (leukemia, monoclonal gammopathy)
- ****Treatment:**** Systemic steroids, immunosuppressants; ****always look for underlying cause****
- > ■ ****Key Insight:**** PG is a ****diagnosis of exclusion**** — rule out infection, vasculitis, and malignancy. The pathergy phenomenon (lesions at sites of trauma) is characteristic.

4. Dermatitis Herpetiformis (DH)

- ****Association:**** ****Celiac disease**** (gluten-sensitive enteropathy)
- ****Presentation:****
- ****Highly pruritic**** (intensely itchy) vesicular rash
- ****Extensor surfaces**** — elbows, knees, buttocks, scalp
- "Button disease" — grouped vesicles
- ****Pathophysiology:**** IgA deposition in dermal papillae
- ****Treatment:**** ****Dapsone**** (rapidly relieves itch), ****gluten-free diet**** (long-term)

> ■ **Quick Fact:** DH is the **skin manifestation of celiac disease**. All patients with DH should be screened for celiac disease, even if asymptomatic GI-wise.

5. Liver Disease — Cutaneous Manifestations

Porphyria Cutanea Tarda (PCT):

- **Mechanism:** Defect in **heme synthesis** (uroporphyrinogen decarboxylase deficiency) → porphyrins accumulate → **photosensitivity**

- **Presentation:**

- Skin fragility, blistering on sun-exposed areas (hands, face)
- Scarring, milia formation
- Hypertrichosis, hyperpigmentation
- **Associations:** Hepatitis C, alcohol, estrogen, iron overload

Xanthomas:

- **Lipid-filled** yellowish deposits in the skin
- Due to **genetic lipid disorders** (familial hypercholesterolemia) or secondary causes (biliary cirrhosis, diabetes)
- Types: Tendinous xanthomas, tuberous xanthomas, xanthelasma, eruptive xanthomas

> ■ **Key Insight:** PCT is the **most common porphyria** and is strongly associated with **Hepatitis C** and **alcohol use**. Always check liver function and iron studies.

■ Hypopigmentation Disorders

| Condition | Inheritance | Key Features |

|---|---|---|

| **Albinism** | AR | Loss of pigment in skin, eyes, hair; **no melanin production**; tyrosinase enzyme defect; eye problems (nystagmus, photophobia) |

| **Piebaldism** | AD | Congenital triangular hypopigmented patch (forelock); disorder of melanocyte development; stable throughout life |

| **Vitiligo** | Acquired | Localized **depigmentation** (not just hypopigmentation); sharply demarcated, symmetrical macular lesions; **loss of melanocytes**; associated with autoimmune diseases (thyroid, DM, Addison's) |

| **Post-inflammatory** | — | After psoriasis, eczema, lichen planus, lupus; infections, chemicals, reactions to pigmented nevi |

> ■ **Key Insight:** **Albinism** = no melanin production (tyrosinase defect); **Vitiligo** = autoimmune destruction of melanocytes; **Piebaldism** = developmental failure of melanocyte

migration.

■ Hyperpigmentation Disorders

Darkening of the skin may be due to:

1. **Increased melanin** (normal pigment) — e.g., Addison's, pregnancy, OCPs
2. **Bile salt deposition** — liver disease (jaundice)
3. **Iron salt deposition** — Haemochromatosis (bronze diabetes)
4. **Drugs or metallic salts** — amiodarone, minocycline, silver, gold
5. **Acanthosis nigricans** — darkening and thickening of axillae, neck, nipples, umbilicus
6. **Post-inflammatory pigmentation** — after eczema, fixed drug eruptions, lichen planus
7. **Malabsorption syndromes** — pellagra (niacin deficiency), scurvy

> ■ **Quick Fact:** Post-inflammatory hyperpigmentation is the **most common cause of hyperpigmentation** in clinical practice, especially in darker skin types.

■ Skin Changes of Underlying Malignancy

Mycosis Fungoides (Cutaneous T-Cell Lymphoma)

- **Type:** Most common primary cutaneous lymphoma (T-cell origin)
- **Clinical presentation varies by stage:**
- **Patch stage:** Flat, scaly, erythematous patches (can mimic eczema)
- **Plaque stage:** Raised, indurated plaques
- **Tumor stage:** Nodular tumors, may ulcerate
- **Erythrodermic stage:** Generalized erythema and scaling (Sézary syndrome)
- **Diagnosis:** Skin biopsy with T-cell clonality studies
- **Treatment:** Topical steroids, phototherapy, chemotherapy, radiation

Poikiloderma

- **Definition:** Combination of **telangiectasia + reticulate pigmentation + atrophy**
- **Causes:**
- **Radiotherapy** (most common iatrogenic cause)
- **Idiopathic**

- **Mycosis fungoides** (poikiloderma vasculare atrophicans)

- Dermatomyositis, lupus

> ■ **Key Insight:** Poikiloderma in an adult should raise suspicion for **mycosis fungoides** — a form of cutaneous T-cell lymphoma.

■ Skin and Pregnancy

| Condition | Timing | Key Features | Effect on Baby |

|---|---|---|---|

| **Prurigo Gestationis** | First trimester (most severe) | Benign, pruritic papular rash; non-specific | None |

| **Polymorphous Eruption of Pregnancy (PEP)** | Third trimester | Pruritic erythematous rash starting in **abdominal striae**; eczematous papules; **spares umbilicus**; no vesicles | None |

| **Pemphigoid Gestationis (PG)** | Second/third trimester | Starts periumbilical; develops **pemphigoid-like vesicles**; spreads to abdomen and thighs; **autoimmune** (anti-BP180 antibodies) | **Small baby, higher mortality; recurrence in future pregnancies** |

> ■ **Mnemonic for PEP vs PG:**

> - **PEP** = **P**oly, **E**czematous, **P**arallel to striae, **P**rotects baby (no effect)

> - **PG** = **P**emphigoid, **G**rave (serious), **G**ets baby (affects fetus), **G**oes again (recurs)

> ■ **Key Insight:** PEP is the **most common** pregnancy dermatosis and is benign. PG is rare but serious — it's an autoimmune blistering disease that can affect the fetus and **recurs with each pregnancy**.

■ Connections & Applications

- **Rash + systemic symptoms** → Always think beyond the skin: vasculitis, autoimmune disease, infection, malignancy

- **Erythema nodosum** is a great example of a **reactive rash** — it's never the primary disease; always investigate

- **Acanthosis nigricans** in a non-obese adult → **malignancy screening** (gastric adenocarcinoma)

- **Dermatitis herpetiformis** → **celiac disease** → screen for gluten-sensitive enteropathy

- **Pemphigoid gestationis** → autoimmune → **fetal monitoring** required

- **Sarcoidosis** → multisystem → check **CXR**, calcium, ACE levels
- **Necrobiosis lipoidica** → check **HbA1c** even if patient is not known diabetic

■ Quick Review Checklist

- ■ Recognize clues to systemic disease from skin findings
- ■ Know the 4 characteristic rashes: EM, Pyoderma gangrenosum, EN, Vasculitis
- ■ Understand toxic erythema (morbilliform, blanching, drug/viral trigger)
- ■ Describe EM: target lesions, HSV trigger, Type IV HSR, minor vs major
- ■ Describe EN: septal panniculitis, shins, Strep/TB/sarcoidosis/IBD causes
- ■ Recognize ECM: Lyme disease, Borrelia, migrating erythema
- ■ Understand sarcoidosis: non-caseating granulomas, BHL, Löfgren syndrome, lupus pernio
- ■ Correlate hormonal skin changes: Addison's, acanthosis nigricans, NLD, thyroid
- ■ Know GI/liver skin manifestations: zinc deficiency, scurvy, PG, DH, PCT, xanthomas
- ■ Differentiate hypopigmentation disorders: albinism, piebaldism, vitiligo
- ■ Understand hyperpigmentation causes: melanin, bile, iron, drugs, post-inflammatory
- ■ Recognize malignancy-associated skin changes: mycosis fungoides, poikiloderma
- ■ Differentiate pregnancy dermatoses: Prurigo, PEP, Pemphigoid gestationis

■ Memory Aids & Mnemonics

| Topic | Mnemonic | Meaning |

|---|---|---|

| **EM target lesion** | **"Doll's Eye"** | Dusky center → pale ring → outer red ring (3 zones) |

| **EN causes** | **"STEPS"** | Strep, TB, Enteric (Yersinia), Pregnancy/OCPs, Sarcoidosis |

| **Löfgren syndrome** | **"FLARE"** | Fever, Lymphadenopathy, Arthritis, Rash (EN), Erythema nodosum |

| **Lyme disease** | **"BORRELIA"** | Bull's eye, Ortho-arthralgia, Remember tick, Recurrent, Erythema migrans, Late arthritis, Immune complex, AV block |

| **Scurvy** | **"SCURVY"** | Swelling gums, Corkscrew hairs, Underperiosteal hemorrhage, Rough skin, Vitamin C deficiency, Yellowing |

| **Addison's pigmentation** | **"ADDISON"** | ACTH Drives Darkening In Skin Of Need |

| **Pregnancy rash** | **"PEP vs PG"** | PEP = Protects baby; PG = Grave, Gets baby, Goes again |

| **Acanthosis nigricans** | **"Dark & Velvety = Insulin or Cancer"** | Obesity/insulin resistance vs. gastric adenocarcinoma |

GOOD LUCK! ■